

Math 115A Final

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- This exam contains 8 pages (including this cover page) and 6 questions for a point total of 80.
- The exam is open book and open notes, but do not discuss the exam with peers or find solutions online.
- Upload a quality scan to Gradescope. Please identify each problem in the scan.

Question	Points	Score
1	10	
2	10	
3	10	
4	20	
5	10	
6	20	
Total:	80	

1. (10 points) Determine whether the statement is true or false. For true statements, provide justification. For false statements, find a counterexample. Each statement is worth 2 points.

(a) The real vector spaces $\mathbb{R}^{n^2}$ and $M_{n \times n}(\mathbb{R})$ are isomorphic.

(b) Let T and U be linear operators on a vector space V . If $U \circ T$ is the zero transformation, then $T \circ U$ is the zero transformation.

(c) If $T : V \rightarrow W$ is linear, then it sends linearly independent sets in V to linearly independent sets in W .

- (d) Let T be a linear operator on a finite dimensional vector space V . Then for any ordered bases $\mathcal{B}$ and $\mathcal{C}$, the matrices $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$ are similar.

- (e) Let T be a linear operator on an n -dimensional vector space V . If T is diagonalizable, then T has n distinct eigenvalues.

2. (10 points) Let $T : V \rightarrow V$ be a linear operator. Prove that $\ker(T) \subset \ker(T^2)$.

3. (10 points) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $T(a_1, a_2) = (a_1 - a_2, a_1, 2a_1 + a_2)$. Let $\mathcal{B}$ be the standard ordered basis for $\mathbb{R}^2$. Let $\mathcal{C} = ((1, 1, 0), (0, 1, 1), (2, 2, 3))$ be a basis for $\mathbb{R}^3$. Compute $[T]_{\mathcal{B}}^{\mathcal{C}}$.

4. (20 points) Each part is worth 10 points. Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be defined as $T(ax^2 + bx + c) = cx^2 + bx + a$.
- (a) Find a basis $\mathcal{B}$ for $P_2(\mathbb{R})$ such that $[T]_{\mathcal{B}}$ is a diagonal matrix.
 - (b) Compute $[T^k]_{\mathcal{C}}$ where $\mathcal{C}$ is the standard ordered basis $\mathcal{C} = (1, x, x^2)$. Note that you will have different answers for odd k and even k .

5. (10 points) Let $T : V \rightarrow V$ be a linear operator on a vector space V . Let $v \in \ker(T)$ be non-zero and w any vector not in $\ker(T)$. Prove that $\{v, w\}$ is linearly independent.

6. (20 points) Each part is worth 10 points. Let W be the subspace of $M_{2 \times 2}(\mathbb{R})$ with basis

$$\mathcal{B} = \left(\begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 9 \\ 5 & -1 \end{pmatrix} \right).$$

Endow the vector space $M_{2 \times 2}(\mathbb{R})$ with the inner product $\langle A, B \rangle = \text{tr}(B^t A)$ where tr is the trace.

- (a) Find an orthonormal basis for W .

- (b) Find the vector in W that is closest to the vector $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in M_{2 \times 2}(\mathbb{R})$ with respect to the inner product.

You do not need to prove that the vector you find is the closest.